

(Please write your Enrollment Number)

Enrollment No. \_\_\_\_\_

**MID-TERM EXAMINATION**

(Course Name: B. Tech.) (Semester: II<sup>nd</sup>)

(March 2025) OFF LINE mode

|                       |                      |
|-----------------------|----------------------|
| Subject Code: BAI 102 | Subject: IT Workshop |
| Time : 1 ½ Hours      | Maximum Marks :30    |

**Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3**

|                |                                                                                                                                                                                                                                    |                   |                   |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|
| <b>Q1</b>      |                                                                                                                                                                                                                                    | <b>(2.5*4=10)</b> | <b>CO Mapping</b> |
|                | a) What are the advantages of R programming?                                                                                                                                                                                       | 2.5               | CO1               |
|                | b) How do we install a package in R Program? What is the difference between a package and a library in R software?                                                                                                                 | 2.5               | CO1               |
|                | c) Write a R Program to create a matrix taking a given vector of numbers as input and define the column and row names. Display the matrix.                                                                                         | 2.5               | CO2               |
|                | d) Define an array. What is the difference between a matrix and an array? Write down the command for creating an array in R.                                                                                                       | 2.5               | CO2               |
| <b>UNIT I</b>  |                                                                                                                                                                                                                                    |                   | <b>CO Mapping</b> |
| <b>Q2</b>      | <b>Attempt any two parts</b>                                                                                                                                                                                                       | <b>(5*2=10)</b>   |                   |
|                | a) i) Write a R program to display the sum of Logarithms of three non-zero values entered by the user.<br>ii) Write a R Program to create a vector. Extract the first and last element of your vector, storing them as new object. | 3+2               | CO1               |
|                | b) Write a program in R :<br>i) To multiply two vectors of different lengths<br>ii) To display the square of a vector                                                                                                              | 2.5+2.5           | CO1               |
|                | c) Write a R Program to find sum, mean and product of a vector.<br>d) Define functions in R. What is the advantage of using functions in R?                                                                                        | 2.5+2.5           | CO1               |
| <b>UNIT II</b> |                                                                                                                                                                                                                                    |                   | <b>CO Mapping</b> |
| <b>Q3</b>      | <b>Attempt any two parts</b>                                                                                                                                                                                                       | <b>(5*2=10)</b>   |                   |
|                | a) Write a program to multiply two matrices in R.                                                                                                                                                                                  | 5                 | CO2               |
|                | b) Write a program in R for inverting a matrix. Verify that the product of these two matrices results in an identity matrix.                                                                                                       | 5                 | CO2               |
|                | c) Create and store a three-dimensional array with six layers of a 4 x 2 matrix, filled with a decreasing sequence of values between 4.8 and 0.1 of the appropriate length                                                         | 5                 | CO2               |

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